IGCSE 0607 Practice Questions — E6.1 — Combined Questions & Markscheme

Units of measure

Total Questions: 12 — **Total Marks:** 84 Marks

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- **E6.1** Units of measure
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

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1 **(a)** Change 52000 cm^2 into m^2.

..... m^2 **[1]**

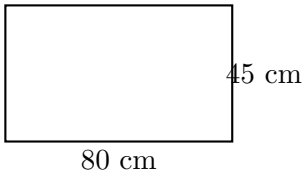
(b) Change 4.7 m^3 into litres.

..... litres **[1]**

(c) Change 850 millilitres into cm^3.

..... cm^3 **[1]**

2 A rectangular piece of card and a water jug are being measured.



(a) Calculate the area of the rectangular piece of card in square metres (m^2).

..... m^2 **[2]**

(b) The jug holds 3.2 litres of water.
Write down this volume in cm^3.

..... cm^3 **[1]**

3 **(a)** Change 3.5 m^2 into cm^2.

..... cm^2 **[1]**

(b) A rectangular tank measures 60 cm by 45 cm by 30 cm.
Calculate the volume of the tank in litres.

..... litres **[3]**

4 A solid wooden block is in the shape of a cuboid.
The dimensions of the block are 40 cm, 250 mm and 0.2 m.

(a) Work out the volume of the block in cm^3.

..... cm^3 **[2]**

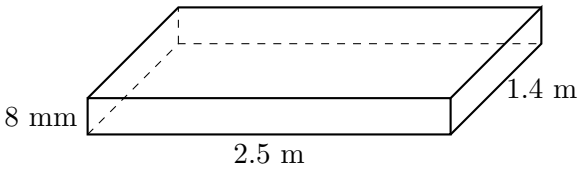
(b) Convert your answer to part (a) into litres.

..... litres **[1]**

(c) The density of the wood is 0.6 g/cm^3.
Calculate the mass of the block in kilograms.

..... kg **[2]**

5 A rectangular block of metal has length 2.5 m, width 1.4 m and thickness 8 mm.



NOT TO SCALE

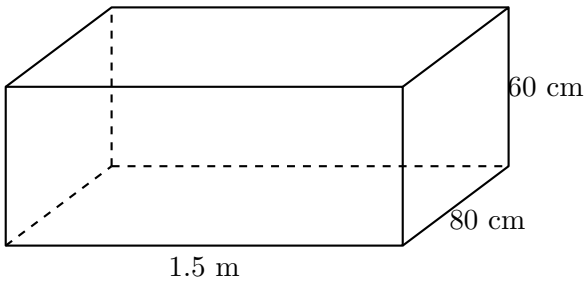
(a) Work out the volume of the block of metal in cm^3.

..... cm^3 **[2]**

(b) The mass of 1 cm^3 of this metal is 7.8 g.
Work out the mass of the block of metal in kilograms.

..... kg **[2]**

6 A metal water tank is made in the shape of a cuboid.



(a) Calculate the capacity of the water tank in litres.

..... litres **[3]**

(b) A solid metal sphere is melted down to make the tank.
The metal sphere has a volume of 0.024 m^3.
The mass of 1 cm^3 of this metal is 7.8 g.

Calculate the total mass of the sphere, giving your answer in kilograms.

..... kg **[3]**

7 During a storm, 15 mm of rain falls on a flat rectangular roof measuring 15 m by 12 m.

(a) Calculate the volume of rain that falls on the roof.
Give your answer in litres.

..... litres **[3]**

(b) The water from the roof flows into a cylindrical tank with a base radius of 60 cm.
Calculate the increase in the depth of the water in the tank.
Give your answer in centimetres, correct to 1 decimal place.

..... cm **[4]**

8 A solid metal sphere has a radius of 15 cm.
The volume, V, of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.

(a) Calculate the volume of the sphere in m^3.
Give your answer in standard form.

..... m^3 **[3]**

(b) The mass of the sphere is 125 kg.
Calculate the density of the metal in g/cm^3.

..... g/cm^3 **[3]**

(c) The sphere is melted down to make completely solid identical cones.
Each cone has a base radius of 24 mm and a height of 5.5 cm.
The volume, V, of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.

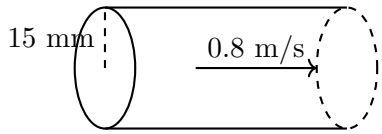
(i) Work out the volume of one cone in cm^3.

..... cm^3 **[3]**

(ii) Work out the maximum number of completely solid cones that can be made.

..... **[2]**

9 Water flows through a cylindrical pipe into an empty rectangular pool.
The pool measures 4 m by 2.5 m.



(a) The cylindrical pipe has a radius of 15 mm.
Water flows through the pipe at a speed of 0.8 m/s.

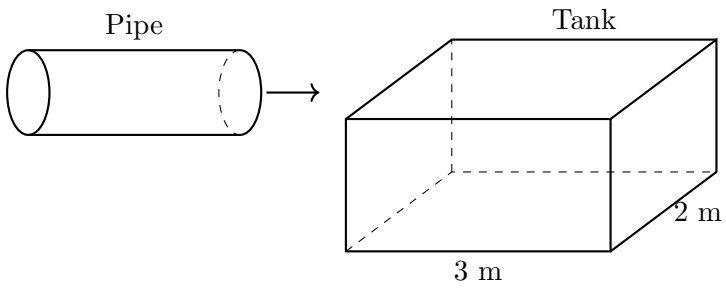
Show that the volume of water flowing out of the pipe in one second is 0.565 litres, correct to 3 significant figures.

[3]

(b) Calculate the time taken for the depth of water in the pool to reach 1.2 m.
Give your answer in hours and minutes, correct to the nearest minute.

..... hours ..... minutes **[4]**

10 Water flows through a cylindrical pipe of internal radius 2.5 cm at a speed of 5.4 km/h.
The water flows into an empty rectangular tank of length 3 m and width 2 m.



NOT TO SCALE

(a) Show that the volume of water flowing into the tank in 1 minute is 176.7 litres, correct to 1 decimal place.

[3]

(b) Calculate the depth of the water in the tank after 45 minutes.
Give your answer in centimetres.

..... cm **[4]**

11 Water flows through a cylindrical pipe of radius 2.5 cm at a speed of 18 km/h.



(a) Calculate the volume of water, in litres, that flows through the pipe in 1 minute.

..... litres **[5]**

(b) The water from the pipe is used to fill an empty rectangular reservoir measuring 50 m by 30 m by 4 m.
Calculate the time it takes to fill the reservoir completely.
Give your answer in days and hours, correct to the nearest hour.

days ..... , hours ..... **[7]**

12 Water flows through a cylindrical pipe of radius 2.5 cm at a speed of 1.6 m/s.

(a) Calculate the volume of water, in litres, that flows through the pipe in one minute.
Give your answer correct to 3 significant figures.

..... litres **[4]**

(b) This water flows into an empty rectangular tank measuring 4.2 m by 3.5 m by 2.4 m.
Calculate the time taken to fill the tank completely.
Give your answer in hours and minutes, correct to the nearest minute.

hours ..... , minutes ..... **[5]**

(c) The outside vertical walls of the rectangular tank are to be painted.
1 litre of paint covers 45000 cm^2.
Paint is sold in 2.5-litre tins.
Work out the minimum number of tins of paint needed.

..... tins **[6]**

1 **(a)** 5.2 **[B1]**

(b) 4700 **[B1]**

(c) 850 **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** $80 \times 45 = 3600$ **[M1]**

0.36 **[A1]**

(b) 3200 **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** 35000 **[B1]**

(b) $60 \times 45 \times 30$ OR 2700×30 OR 81000 **[M1]**

their $81000 \div 1000$ **[M1]**

81 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Converts units correctly to cm: 250 mm = 25 cm AND 0.2 m = 20 cm **[B1]**

20000 **[A1]**

(b) 20 **[B1]**

(c) Uses Mass = Density × Volume: 0.6×20000 (= 12000 g) **[M1]**

12 **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** Converts all dimensions to cm: 250, 140, 0.8 **[M1]**

$250 \times 140 \times 0.8 = 28000$ **[A1]**

(b) Multiplies volume by 7.8: $28000 \times 7.8 = 218400$ **[M1]**

218.4 **[A1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** Converts dimensions to consistent units, e.g. 150 cm, 80 cm, 60 cm OR 1.5 m, 0.8 m, 0.6 m **[M1]**

$150 \times 80 \times 60 = 720000$ OR $1.5 \times 0.8 \times 0.6 = 0.72$ **[M1]**

720 **[A1]**

(b) $0.024 \times 1000000 = 24000$ **[M1]**

Mass $= 24000 \times 7.8 = 187200$ **[M1]**

187.2 **[A1]**

[Back to Question] *[Total Marks: 6]*

7 **(a)** Conversion of units: 15 $mm = 0.015$ m OR 15 $m = 1500$ cm etc. **[M1]**

Volume $= 15 \times 12 \times 0.015$ (or eq) **[M1]**

2700 **[A1]**

(b) Use of consistent units for tank volume equation (e.g. $r = 0.6$ m and $V = 2.7$ m^3 OR $V = 2700000$ cm^3 and $r = 60$ cm) **[M1]**

$\pi \times 0.6^2 \times h = 2.7$ OR $\pi \times 60^2 \times h = 2700000$ **[M1]**

$h = \dfrac{2.7}{0.36\pi}$ (m) OR $h = \dfrac{2700000}{3600\pi}$ (cm) **[M1]**

238.7 **[A1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** $\dfrac{4}{3} \times \pi \times 15^3$ OR 4500π OR 14137 **[M1]**

their $14137 \div 1000000$ OR $\dfrac{4}{3} \times \pi \times 0.15^3$ **[M1]**

1.41×10^{-2} **[A1]**

(b) 125 kg = 125000 g **[B1]**

$\dfrac{125000}{4500\pi}$ OR $\dfrac{125000}{14137}$ **[M1]**

8.84 (8.841 . . .) **[A1]**

(c) **(i)** 24 mm = 2.4 cm **[B1]**

$\dfrac{1}{3} \times \pi \times 2.4^2 \times 5.5$ **[M1]**

33.2 (33.17 . . .) OR 10.56π **[A1]**

(ii) $\dfrac{4500\pi}{\text{their } 33.17}$ OR 426.1 . . . **[M1]**

426 **[A1]**

[Back to Question] *[Total Marks: 11]*

9 **(a)** Identifies consistent units for radius and length (e.g. $r = 1.5$ cm and length = 80 cm) **[M1]**

Volume in one second $= \pi \times 1.5^2 \times 80 = 180\pi$ cm^3 $(= 565.48 \ldots$ $cm^3)$ **[M1]**

Converts to litres $(180\pi \div 1000)$ and confirms $0.56548 \cdots \approx 0.565$ **[A1]**

(b) Volume of pool required $= 4 \times 2.5 \times 1.2 = 12$ m^3 OR 12000 litres **[M1]**

Time in seconds $= 12000 \div 0.56548 \ldots$ OR $12000 \div \dfrac{180\pi}{1000}$ **[M1]**

$= 21220.6 \ldots$ seconds (353.6 . . . minutes) **[A1]**

5 hours 54 minutes **[A1]**

[Back to Question] *[Total Marks: 7]*

10 (a) Converts speed to cm/min: $5.4 \times 1000 \times 100 \div 60 = 9000$ **[M1]**

Calculates volume in 1 minute: $\pi \times 2.5^2 \times 9000 = 56250\pi\ cm^3$ **[M1]**

Converts cm^3 to litres and evaluates: $56250\pi \div 1000 = 176.714\ldots$ **[A1]**

(b) Calculates total volume in 45 minutes: $56250\pi \times 45 = 2531250\pi\ cm^3$ (or uses litres: $176.714\cdots \times 45 = 7952.1\ldots$ litres) **[M1]**

Converts tank base dimensions to cm and finds area: $300 \times 200 = 60000\ cm^2$ **[M1]**

Sets up equation for depth: Depth $= \dfrac{2531250\pi}{60000}$ **[M1]**

133 (or 132.5 to 132.6) **[A1]**

[Back to Question] *[Total Marks: 7]*

11 (a) Cross-sectional area $= \pi \times 2.5^2$ $(= 6.25\pi)$ **[M1]**

Speed conversion: $18\ km/h = \dfrac{18 \times 1000 \times 100}{60}\ cm/min$ $(= 30000\ cm/min)$ **[M1]**

Volume per minute $= 6.25\pi \times 30000$ $(= 187500\pi\ cm^3)$ **[M1]**

Convert to litres: $\dfrac{187500\pi}{1000}$ **[M1]**

589 or 589.0 to 589.1 (187.5π) **[A1]**

(b) Capacity of reservoir $= 50 \times 30 \times 4 = 6000\ m^3$ OR 6000000 litres **[M1]**

Time in minutes $= \dfrac{6000000}{187.5\pi}$ (or 6000000÷ their (a)) $(= 10185.9...)$ **[M1]**

Time in hours $= \dfrac{10185.9...}{60}$ $(= 169.76...)$ **[M1]**

$169.76... \div 24 = 7.07...$ days so 7 full days **[M1]**

$0.07... \times 24 = 1.76...$ hours (or $169.76... - 7 \times 24$) **[M1]**

7 **[A1]**

2 **[A1]**

[Back to Question] *[Total Marks: 12]*

12 (a) $\pi \times 2.5^2 \times 160$ OR $\pi \times 0.025^2 \times 1.6$ **[M1]**

$\times 60$ **[M1]**

$\div 1000$ (if using cm) OR $\times 1000$ (if using m) **[M1]**

188 $(188.4\ldots)$ **[A1]**

(b) Volume of tank $= 4.2 \times 3.5 \times 2.4 = 35.28\ m^3$ **[M1]**

$35.28 \times 1000 = 35280$ litres **[M1]**

Time $= \dfrac{35280}{\text{their } 188.49}$ OR $\dfrac{35280}{60\pi}$ **[M1]**

187 minutes (187.1 . . .) **[A1]**

3 hours 7 minutes **[A1]**

(c) Calculates lateral area correctly: $2 \times 4.2 \times 2.4 + 2 \times 3.5 \times 2.4$ OR $36.96\ m^2$ [M2]
(If incorrect, M1 for one correct face area e.g., 4.2×2.4)

$36.96 \times 10000 = 369600\ cm^2$ OR $45000 \div 10000 = 4.5\ m^2$ **[M1]**

Litres of paint required $= \dfrac{369600}{45000}$ OR $\dfrac{36.96}{4.5}$ **[M1]**

Number of tins $= \dfrac{8.21\ldots}{2.5}$ $(= 3.28\ldots)$ **[M1]**

4 **[A1]**

[Back to Question] *[Total Marks: 15]*